NGR Lowline — CTO v1.3

Dispatcher Serial Commands Reference

Firmware v1.3 | Packet VERSION 6 | April 2026

All commands typed into the **Dispatcher** serial monitor at 115200 baud. Press Enter. Case-insensitive. Changes take effect immediately and reset to compiled defaults on power cycle.

MOTION

Command	Target	Action
`g`	Both	GO — start both locos at cruise PWM
`s`	Both	STOP — pause both (retains encounter state)
`e`	Both	E-STOP — immediate halt, clears all state. Power cycle locos to restart.
`og`	Otto	GO Otto only
`tg`	Toby	GO Toby only
`os`	Otto	STOP Otto only
`ts`	Toby	STOP Toby only

DIAGNOSTICS

Command	Action
`pt`	Print peer/audit table — last known block, seq, age for each loco
`c`	CLEAR ALL — reset peer tables and CTO encounter state on all locos

CRUISE PWM (0-255, per locomotive)

Command	Target	Example	Action
`op###`	Otto	op110	Set Otto cruise PWM to 110
`tp###`	Toby	tp120	Set Toby cruise PWM to 120
`bp###`	Both	bp115	Set both locos cruise PWM to 115

Each loco may need a different cruise PWM. Otto and Toby can run at different speeds.

RAMP RATES (ms per PWM step — higher = slower/gentler)

Command	Scope	Example	Action
`rr###`	All profiles	rr80	Shared decel rate — all roles simultaneously. Default 80ms/step.
`ru###`	All departures	ru120	Shared accel rate — all departures from all stops. Default 120ms/step.

rr53 = firm trailing brake. rr80 = normal. rr150 = very gentle. ru50 = fast launch. ru120 = smooth. ru200 = very slow pull-away.

ROLE PROFILE PARAMETERS (ramp delay and dwell)

Format: **[target][role][param][value]** — no spaces.

Target	Loco
`o`	Otto only
`t`	Toby only
`b`	Both locos

Role	Profile
`df`	DEFAULT (startup/unknown)
`st`	STATION (normal stop)
`tr`	TRAILING (closing on leader)
`ld`	LEADER (holding for trailer)

Param	Description
`rd`	Ramp delay ms (coast before braking)
`dw`	Dwell ms (stopped before departure gate)

Example	Effect
`bldrd6000`	Both: LEADER ramp delay = 6000ms (6s coast to platform)
`oldrd6000`	Otto: LEADER ramp delay = 6000ms
`bstdw5000`	Both: STATION dwell = 5000ms
`btrdw500`	Both: TRAILING dwell = 500ms
`bdfrd200`	Both: DEFAULT ramp delay = 200ms

TRAILING PLATFORM SEQUENCE PARAMETERS (ms, up to 30000)

Format: **[target][2-letter code][value]**

Command	Code	Default	Description
`[o|t|b]fd#####`	fd	3000ms	Follow delay — wait after DEPARTING before advancing
`[o|t|b]ad#####`	ad	10000ms	Advance duration — time moving toward platform at advance PWM
`[o|t|b]ap###`	ap	cruise/2	Advance PWM — speed during approach. 0 = auto (cruisePwm/2)
`[o|t|b]pd#####`	pd	10000ms	Platform dwell — time stopped at platform (boarding simulation)

Example	Effect
`bfd3000`	Both: follow delay = 3000ms
`bad10000`	Both: advance duration = 10000ms
`bap60`	Both: advance PWM = 60 (fixed)
`bap0`	Both: advance PWM = auto (cruisePwm/2)
`bpd10000`	Both: platform dwell = 10000ms

advanceMs compensates for leader's 6s coast at half speed. Start 10000ms, tune up if trailing stops short of platform.

QUICK REFERENCE

g	s	e	c	pt
GO both	STOP both	E-STOP	CLEAR ALL	Peer table
rr###	**ru###**	**bp###**	**bldrd####**	**bad#####**
Decel rate	Accel rate	Both PWM	Leader coast	Advance time

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Field Test Checkoff Sheet

Date: ____________________ Session: ____________________ Weather: ____________________

PRE-RUN VERIFICATION

- ■ All three ESPs flashed with v1.3 (VERSION=6) — Otto, Toby, Dispatcher
- ■ Dispatcher serial monitor open at 115200 baud
- ■ Startup banners confirmed on both locos — correct version and LOCO_ID
- ■ Hall calibration completed on both locos — baseline and thresholds printed
- ■ Track clear, magnets secure at all four stations (PATIO / BAMBOO / ARCHES / GRILLERS)
- ■ Both loco batteries charged and power cars connected

INITIAL STARTUP

- ■ Type 'g' in Dispatcher — both locos receive CMD_START
- ■ Both locos ramp up slowly (120ms/step accel) — no lurch
- ■ Both locos reach cruise PWM and run steadily
- ■ Dispatcher log shows RX BLOCK from both locos within first lap

STATION STOP — NO ENCOUNTER (trains well separated)

- ■ Loco hits block — Dispatcher: RX BLOCK
- ■ Loco serial: RAMP DELAY expired — braking (after rampDelayMs coast)
- ■ Loco serial: CONFIRMED STOP
- ■ Loco dwells, then: DWELL COMPLETE -> RUNNING
- ■ Dispatcher: RX DEPARTING from loco
- ■ Loco resumes — slow ramp-up observed physically

ENCOUNTER DETECTION

- ■ Trailing loco arrives at block where leader recently stopped
- ■ Dispatcher: RX ENC_START trailing=LOCO leader=LOCO block=BLOCK
- ■ Dispatcher: RX RELEASE trailing=LOCO leader=LOCO block=BLOCK
- ■ Leader serial: LEADER: RELEASE accepted
- ■ Leader departs — Dispatcher: RX DEPARTING sender=LEADER block=BLOCK

TRAILING PLATFORM ADVANCE SEQUENCE — CRITICAL PATH

- ■ Trailing stops behind leader — CONFIRMED STOP before RELEASE sent
- ■ Trailing serial: waiting for DEPARTING from [leaderID]

- ■ DEPARTING received — trailing serial: DEPARTING accepted — leader moving
- ■ Trailing serial: follow delay 3000ms

Follow delay expires:

- ■ Trailing serial: advancing to platform for 10000ms
- ■ Trailing loco physically moves at reduced speed (cruise/2)
- ■ Advance covers approximately the leader's 6s coast distance

Advance timer expires:

- ■ Trailing serial: advance complete — platform stop
- ■ Trailing loco brakes — CONFIRMED STOP
- ■ Trailing serial: platform dwell 10000ms

Platform dwell complete:

- ■ Trailing serial: platform dwell complete — resuming
- ■ Trailing ramps up and resumes normal running
- ■ Dispatcher: RX DEPARTING from trailing loco

SAFETY CHECKS — MUST ALL PASS

- ■ NO rear-end collision at any block during encounter
- ■ Leader physically moving BEFORE trailing loco begins advancing
- ■ Trailing does NOT depart on dwell expiry alone — waits for DEPARTING
- ■ RELEASE never sent while trailing loco visibly still moving
- ■ Both locos NOT departing simultaneously (phase-lock prevention)

RUNTIME TUNING LOG

Time	Command sent	Observed result / notes

ISSUES / OBSERVATIONS

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SESSION RESULT

■ All critical path checks passed — system operating correctly

■ Issues noted above — carry forward to next session

Final parameter values this session:

Parameter	Otto	Toby	Both
Cruise PWM			—
Leader rampDelayMs			—
rampRateMs (decel)	—	—	
rampUpRateMs	—	—	
followDelayMs	—	—	
advanceMs	—	—	
advancePwm	—	—	
platformDwellMs	—	—	